

StaySense BI

AI-Powered Guest Sentiment Intelligence for Homestays

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Business Challenge

Homestay owners often struggle to make informed decisions about pricing, room offerings, and occupancy due to fragmented data. StaySense BI transforms Airbnb listing data into actionable insights, helping hosts optimize revenue and guest experience.

Speaker Notes: Welcome. Today I'm presenting StaySense BI, an AI-powered solution for homestay hosts. Currently, owners struggle with massive, unorganized data, leading to missed opportunities. StaySense simplifies this, turning raw Airbnb data into actionable revenue and occupancy strategies.

Dataset Overview



Listings

48,895



Boroughs

5



Room Types

3



Price Categories

4

Features Used: Borough, Room Type, Price, Number of Reviews, Availability, Minimum Nights

Analysis Tool: Python (Pandas & Matplotlib), Looker Studio Dashboard

Speaker Notes: We utilized the 2019 NYC Airbnb dataset with nearly 50,000 listings across 5 boroughs. Our tech stack leveraged Python for cleaning and analysis, combined with Looker Studio for visual reporting, enabling us to quickly identify trends in pricing and room type engagement.

Key Finding: Average Pricing

My analysis confirms Manhattan commands a significant premium in listing prices compared to other boroughs.

Manhattan has the highest average listing price (**\$196.88**), significantly exceeding Brooklyn (**\$124.38**) and the remaining boroughs. This indicates a stronger revenue opportunity for premium accommodations.

Average Manhattan Listing Price

\$196.88



Speaker Notes: The data is clear: location drives pricing. Manhattan listings average nearly \$200 per night, indicating high revenue potential. This insight helps us segment our strategy; Manhattan listings require a high-touch approach, whereas other boroughs may prioritize volume.

Business Impact



Revenue: Higher pricing opportunities



Guest Engagement: Entire Home listings receive the highest reviews



Decision Support: BI dashboards enable continuous monitoring

Data Source: Airbnb Open Data (2019)

Dashboard created using Google Looker Studio

Made by: Juhi Kansal

<https://datastudio.google.com/s/rX9SHra7o4w>

Speaker Notes: Our findings show that "Entire Home" listings are the clear winners in guest engagement. We also discovered that availability alone isn't a proxy for success. Moving forward, hosts should focus on strategic positioning and borough-specific pricing to drive growth.

Strategic Recommendations

 **Optimize Pricing Strategy:** Implement borough-specific pricing models to reflect local market demand and maximize revenue potential.

 **Focus on High-Performing Listings:** Promote and optimize **Entire Home/Apartments**, as they generated the highest guest engagement in the dataset.

 **Enable Continuous Decision-Making:** Leverage the StaySense BI dashboard to monitor pricing trends, customer reviews, and listing performance for data-driven decisions.

Key Takeaway: *Combining Business Intelligence with AI-assisted analysis enables homestay owners to make faster, smarter, and evidence-based business decisions.*